

BRANDON CHEN

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EDUCATION

New York City College of Technology

B.S. in Mechanical Engineering

Expected May 2027

GPA: 3.6/4.00

RELEVANT COURSEWORK

Materials Science, Strength of Materials, Engineering Graphics, Manufacturing Processes, Thermodynamics, Kinematics/Dynamics of Machines, Materials Testing Lab, Embedded Systems & Robotics, Advanced Solids Modeling, Quality Control

TECHNICAL SKILLS

- **Mechanical:** GD&T (ASME Y14.5), tolerance stack-up analysis, stress/strain analysis, shear & bending moment diagrams, beam deflection calculations
- **Robotics/Embedded:** Arduino UNO R4 Minima, Raspberry Pi, ESP8266 WiFi, ultrasonic/IR sensors, PWM motor control, H-bridge drivers, serial communication
- **CAD/CAM:** SolidWorks, AutoCAD, Autodesk Inventor, Mastercam (2D contour, pocketing, drilling, facing), CNC fundamentals, 3D printing
- **Programming:** C/C++ (Arduino), Python (Raspberry Pi), MATLAB
- **Fabrication:** Precision riveting, welding, heat treatment, electrical troubleshooting, micrometers, torque wrenches

CERTIFICATIONS

- **Basic CNC Programming and CAM Fundamentals**, SolidProfessor / Mastercam Training (2025)

WORK EXPERIENCE

NYC Health + Hospitals/Woodhull

Engineering Intern

Jul 2023 – Aug 2023

- Performed preventative maintenance and corrective repairs on electromechanical medical systems including EKG, ultrasound, and radiology equipment.
- Diagnosed electrical and mechanical faults using multimeters, circuit analysis, and structured troubleshooting methods.
- Assisted in precision disassembly, alignment verification, and reassembly while ensuring functional integrity and compliance.
- Documented root-cause findings and corrective actions in technical service reports to support equipment reliability.
- Evaluated equipment condition and prioritized servicing based on operational risk and performance impact.

PROJECTS

Arduino Robotics & IoT Systems Development

Embedded Systems & Robotics | 2026

- Programmed Arduino UNO R4 Minima microcontroller (C/C++) for sensor integration and closed-loop PWM motor control.
- Developed smart car navigation system using ultrasonic and infrared obstacle detection with real-time feedback logic.
- Integrated ESP8266 WiFi module to enable wireless communication and basic IoT functionality.
- Designed and assembled mechanical chassis components while debugging hardware-software integration.

Structural Design of Pedestrian Walkway Beam System

Mechanical Engineering Design Project | 2024

- Modeled multi-load scenarios including pedestrian and wind loading to determine worst-case structural demand.
- Constructed shear force and bending moment diagrams to locate maximum internal stresses and governing failure points.
- Performed stress and deflection analysis using elastic beam theory and verified compliance with serviceability limits.
- Conducted material trade study comparing structural steel and reinforced concrete based on stiffness and durability.

Advanced Parametric Design, CAM Programming & CNC Machining

AutoCAD / Mastercam | 2025

- Developed fully constrained parametric 3D CAD models using design intent principles, feature-based modeling, and tolerance-driven dimensioning.
- Translated 3D geometry into manufacturing strategy within Mastercam, generating 2D contour, pocketing, drilling, and facing toolpaths.
- Selected tooling and optimized cutting parameters to balance material removal rate, surface finish, and tool wear.
- Post-processed toolpaths to generate G-code and executed machining operations on a CNC vertical mill, verifying dimensional accuracy.

Aircraft Structural Fabrication & Engine Maintenance

Aerospace Manufacturing & Maintenance | 2019

- Fabricated composite aileron structures using precision riveting, epoxy bonding, welding, and controlled heat treatment processes.
- Maintained tight geometric tolerances in accordance with aerospace manufacturing standards.
- Disassembled and inspected reciprocating aircraft engines for wear and dimensional compliance.
- Utilized precision measurement tools including micrometers and torque wrenches during repair and reassembly.
- Fabricated and pressure-tested aircraft fluid lines in compliance with aerospace safety standards.